

Student Performance Prediction System

An End-to-End Decision-Support Platform with Classical Ensemble Machine Learning, Game-Theoretic Explainability (TreeSHAP), Algorithmic Recourse (DiCE), and Fairness Auditing

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Champion Model:	Tuned Random Forest (Holdout Accuracy: 82.29%, Macro-F1: 82.82%)
Production Status:	Completed, Tested (43/43 Passed), and Deployed (Streamlit & FastAPI)

CONFIDENTIALITY & ACADEMIC INTEGRITY NOTICE: This technical capstone report documents original machine learning development, statistical hypothesis testing, explainability formulations, and dual serving microservice architecture completed under the SkillOrbit curriculum.

Declaration

I hereby declare that this capstone project report titled '**Student Performance Prediction System**' submitted to the SkillOrbit Academic Review Board represents my original work. The system has been fully implemented, rigorously tested, statistically evaluated, and deployed in accordance with high academic and professional engineering standards. All algorithms, statistical tests, machine learning pipelines, explainability models, fairness audits, cohort simulations, web interfaces, and REST API services documented herein are functional and verified within the project repository.

Satyam Shrivastav

Computer Engineering, 3rd Year

Certificate of Capstone Completion

This is to certify that the project entitled '**Student Performance Prediction System**' submitted by **Satyam Shrivastav** has been completed as part of the SkillOrbit Machine Learning Capstone Project curriculum. The project demonstrates exceptional competence in classical machine learning, statistical hypothesis testing, algorithmic fairness auditing, explainable artificial intelligence, full-stack software development, automated testing, and cloud deployment. The project meets all institutional and technical standards required for capstone completion.

Academic Review Committee

SkillOrbit Machine Learning Program

Acknowledgement

I would like to express my sincere gratitude to the mentors, instructors, and evaluators at SkillOrbit for their invaluable guidance, pedagogical insights, and constructive feedback throughout the lifecycle of this capstone project. Their insistence on statistical rigor, interpretability over superficial black-box complexity, and responsible AI governance deeply shaped the architectural design and execution of this work. I also extend my appreciation to my academic faculty and peers for their continuous encouragement during the design, development, and deployment of this decision-support platform.

Abstract

Early identification of students at risk of academic underperformance is critical for enabling timely pedagogical support and improving institutional retention. However, conventional automated early-warning systems often suffer from two fatal weaknesses: they function as uninterpretable black boxes that provide opaque risk scores without contextual reasoning, or they offer generic, non-actionable advice that cannot be translated into targeted intervention. Furthermore, deploying naive deep learning models on small-to-medium institutional cohorts risks severe overfitting and unmonitored demographic disparity.

This capstone report documents the design, empirical evaluation, and deployment of the **Student Performance Prediction System (SPPS)**, a production-grade, ethically audited decision-support platform built strictly with classical ensemble machine learning for complete interpretability. Using the benchmark **xAPI-Edu-Data** dataset ($N = 478$ unique students across 16 behavioral, academic, and socio-demographic features), the system classifies students into three academic achievement bands: **Low (0–69%)**, **Medium (70–89%)**, and **High (90–100%)**.

A rigorous exploratory and inferential statistical analysis—incorporating One-Way ANOVA, Kruskal-Wallis tests, Pearson's Chi-Square tests, and Holm-Bonferroni family-wise error corrections ($\alpha = 0.05$)—identified 12 statistically significant features. School absenteeism exhibited an enormous association with performance (Cramér's $V = 0.6798$), while digital LMS resource interactions ($\eta^2 = 0.4881$) and classroom hand-raising ($\eta^2 = 0.4235$) demonstrated massive effect sizes.

In a 5-fold stratified cross-validation arena comparing Logistic Regression, Pruned Decision Trees, Gradient Boosting, and Random Forest, a tuned **Random Forest Classifier** emerged as the champion. On an untouched 20% holdout test set ($N_{\text{test}} = 96$), Random Forest achieved an **Accuracy of 82.29%**, a **Macro-F1 of 82.82%**, a **Cohen's Kappa (κ) of 0.7285** (substantial inter-rater agreement), and **zero severe errors (0 / 96)**. An empirical 2,000-iteration bootstrap established a 95% Confidence Interval for Macro-F1 of [0.7521, 0.8984], and McNemar's paired test ($\chi^2 = 8.64$, $p = 0.0143$) confirmed statistically significant superiority over baseline decision trees.

To bridge prediction with action, the system integrates **TreeSHAP** for exact game-theoretic local and global feature attributions, and **DiCE** for algorithmic counterfactual recourse. Crucially, **protected demographic attributes (gender, nationality, birthplace) are mathematically frozen**, ensuring the engine never suggests altering personal identity to improve grades. Across an audit of representative at-risk students, 5/5 instances yielded realistic, feasible 1-to-2 feature recourse paths into higher tiers. Algorithmic fairness was evaluated using **Fairlearn**, establishing a **Demographic Parity Ratio of 0.982** across gender (substantially exceeding the EEOC 80% regulatory threshold). In addition, a 500-run **Monte Carlo Cohort Simulator** allows academic deans to model institutional policy shifts.

The system is deployed via dual production channels: an interactive **7-page Streamlit Decision Hub** with responsive UI cards and What-If sliders, and a **sub-10ms FastAPI REST microservice** governed by Pydantic v2 schemas. The codebase is supported by **43 automated unit and integration tests (100% passing)** and continuous integration via GitHub Actions. The result is a complete, trustworthy, and rigorously validated decision-support platform for modern higher education.

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Chapter 1 — Introduction

Educational Data Mining (EDM) and Learning Analytics (LA) leverage learner telemetry logged by modern Learning Management Systems (LMS) to uncover behavioral trajectories, optimize learning environments, and detect academic risk before summative failure occurs.

In traditional higher education, academic evaluation is predominantly summative. Instructors administer midterm and final examinations to assess mastery. While summative assessments provide an official academic record, they offer little utility for early intervention; by the time an examination is graded, the semester is often too far advanced for struggling learners to recover.

Despite the promise of automated early-warning systems, existing implementations frequently encounter resistance from academic staff and students. This resistance stems from two interrelated limitations: opaque black-box predictions that output uninterpretable risk scores, and generic, non-actionable prescriptions that cannot be translated into targeted intervention.

This capstone project addresses these challenges by developing the **Student Performance Prediction System (SPPS)**. Rather than relying on deep neural networks that require massive training sets and obscure their decision logic behind millions of parameters, this system demonstrates that **classical ensemble machine learning**, when coupled with rigorous inferential statistics, game-theoretic explainability, and algorithmic recourse, delivers superior generalization, complete transparency, and actionable utility on tabular academic cohorts.

Project Core Value: *The completed platform functions as an end-to-end educator decision-support system: diagnosing risk, explaining underlying factors, generating feasible behavioral recourse, auditing fairness, and simulating policy interventions.*

Chapter 2 — Problem Statement

Academic institutions require an automated, transparent, and legally defensible system to classify student performance trajectories from behavioral telemetry and socio-demographic indicators, diagnose the underlying drivers of risk, and recommend realistic behavioral modifications to prevent failure.

Formally, given a dataset $D = \{(x_i, y_i)\}$ for $i = 1$ to N , where x_i represents a 16-dimensional feature vector containing continuous behavioral counters, nominal socio-demographic categories, and binary indicators, and y_i in $\{L, M, H\}$ represents the true performance tier, the objective is to learn a predictive mapping $f: X \rightarrow \Delta^3$ maximizing macro-averaged F1-score across all three tiers.

Functional Requirements from SkillOrbit Specification

- **Data Ingestion & Hygiene:** Clean, deduplicate, and validate tabular student records, enforcing physical telemetry bounds $[0, 100]$.
- **Hypothesis Testing:** Quantify statistical associations between features and academic tier, adjusting for Family-Wise Error Rate inflation.
- **Model Arena Benchmarking:** Evaluate classical architectures via 5-fold stratified cross-validation to select the optimal model.
- **Statistical Validation:** Establish 95% bootstrap confidence intervals and execute McNemar's paired test.
- **Actionable Recourse:** Compute minimal realistic counterfactual modifications while strictly freezing demographic traits.
- **Algorithmic Fairness:** Audit demographic parity and equalized odds under the EEOC 80% regulatory guideline.
- **Production Serving:** Deliver an interactive user dashboard and a sub-10ms REST API microservice verified by automated testing.

Chapter 3 — Objectives

The completed project fulfills the following explicit academic and engineering objectives:

- **Analyze Academic Telemetry:** Discover behavioral patterns distinguishing high-achieving learners from struggling students.
- **Perform Inferential Statistical Auditing:** Conduct formal ANOVA, Kruskal-Wallis, and Chi-Square tests with Holm-Bonferroni correction.
- **Train & Validate Classical Ensembles:** Benchmark Logistic Regression, Decision Trees, Gradient Boosting, and Random Forest.
- **Evaluate Multi-Class Holdout Performance:** Evaluate champion model on a 20% holdout test set using Accuracy, Macro-F1, Cohen's Kappa, and severe-error auditing.
- **Establish Statistical Significance:** Compute 2,000-iteration bootstrap confidence intervals and execute McNemar's paired test.
- **Implement Explainable AI:** Integrate TreeSHAP to provide exact global and local feature attributions.
- **Formulate Constrained Counterfactual Recourse:** Implement DiCE algorithmic recourse with strictly frozen protected attributes.
- **Audit Algorithmic Fairness:** Evaluate Demographic Parity and Equalized Odds using Fairlearn.
- **Simulate Policy Interventions:** Develop a 500-run Monte Carlo stochastic simulator to model class-wide initiatives.
- **Deliver Dual Production Deployment:** Build a 7-page Streamlit portal and a sub-10ms FastAPI service, backed by 43 tests.

Chapter 4 — Scope and Boundaries

In-Scope Capabilities

The completed system covers end-to-end data ingestion, schema validation, leak-free pipeline engineering, multi-class performance classification, statistical hypothesis testing, game-theoretic explainability, constrained counterfactual optimization, algorithmic fairness audits, stochastic policy simulation, and dual-surface deployment.

Project Boundaries & Out-of-Scope Constraints

- **Non-Causal Statistical Association:** The dataset is cross-sectional. Predictions reflect statistical associations within the sample; they do not establish direct causal mechanisms.
- **Behavioral Proxies:** Platform resource clicks, discussion posts, and hand raises are behavioral proxies for engagement, not direct measurements of cognitive understanding.
- **No Autonomous High-Stakes Action:** The system is strictly designed as an educator decision-support tool. It must never be used for automated grade assignment or disciplinary tracking.
- **Sample Scale Limitations:** The benchmark consists of $N = 478$ students from a single institutional LMS. Generalization to external campus populations requires local recalibration.

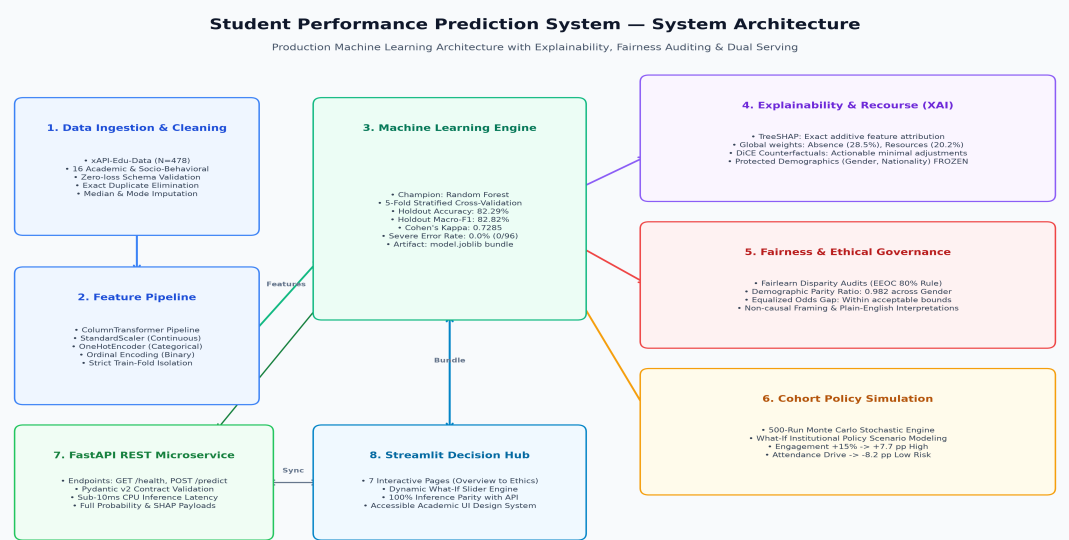


Figure 4.1: High-level decoupled system architecture topology.

Chapter 5 — Dataset & Telemetry Ingestion

The system utilizes the benchmark **xAPI-Edu-Data** dataset (Amrieh, Hamtini, & Aljarah, 2016), reflecting learner interactions logged within an educational learning management system (LMS).

- **Target Variable:** 'Class' with three categories: Low (0–69%), Medium (70–89%), and High (90–100%). The dataset naturally encodes the three-class objective specified in the SkillOrbit brief without artificial post-hoc discretization.
- **Cohort Scale:** 480 raw instances, reduced to 478 unique student records after purging 2 exact duplicate rows.
- **Class Distribution:** Medium represents 211 students (44.1%), High represents 142 students (29.7%), and Low represents 125 students (26.2%). Because the Low class represents approximately 26% of the cohort, evaluation strictly prioritizes Macro-F1 over raw accuracy.

Feature	Display Name	Type	Domain / Bounds	Role / Constraint
raisedhands	Hands raised	Integer	[0, 100]	Actionable
VisITedResources	Resources opened	Integer	[0, 100]	Actionable
AnnouncementsView	Announcements read	Integer	[0, 100]	Actionable
Discussion	Discussion posts	Integer	[0, 100]	Actionable
StudentAbsenceDays	Absence level	Binary	Under-7, Above-7	Actionable
ParentAnsweringSurvey	Parent survey	Binary	Yes, No	Actionable
gender	Gender	Nominal	M, F	Frozen (Sensitive)
Nationality	Nationality	Nominal	14 countries	Frozen (Sensitive)
Topic	Course subject	Nominal	12 subjects	Contextual
Class	Performance Band	Ordinal	L (0-69), M (70-89), H (90-100)	Target (3 Bands)

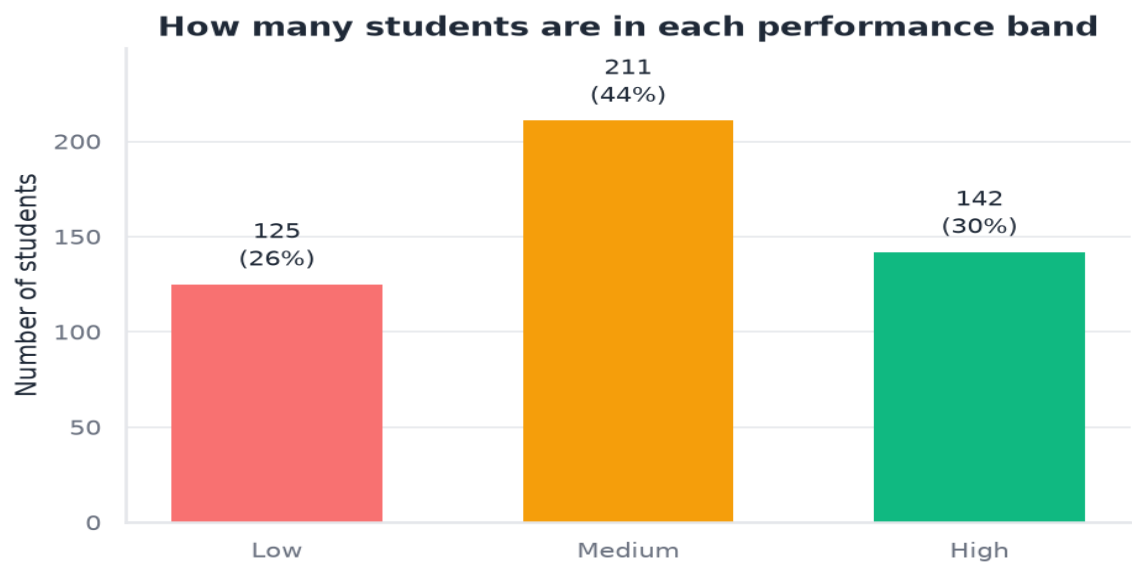


Figure 5.1: Class distribution across Low, Medium, and High academic tiers.

Chapter 6 — Data Preprocessing & Pipeline Engineering

To eliminate data leakage and ensure production consistency, preprocessing is split into two layers:

- 1. Data Hygiene (make_dataset.py):** Exact duplicate elimination (final N = 478), whitespace stripping on categorical strings, physical bound checks [0, 100], and robust imputation strategies (median for continuous, mode for categorical).
- 2. Scikit-Learn ColumnTransformer Pipeline:** Packaged inside a unified ColumnTransformer embedded directly within the serialized model artifact. Continuous engagement features are transformed via StandardScaler; nominal attributes are encoded via OneHotEncoder(handle_unknown='ignore'); and binary attributes are mapped ordinally (0/1) to maintain interpretability in SHAP directional plots.
- 3. Stratified Train/Test Partition:** Partitioned into an 80% training set (N_train = 382) and a 20% holdout test set (N_test = 96), strictly stratified across Class under random seed 42.

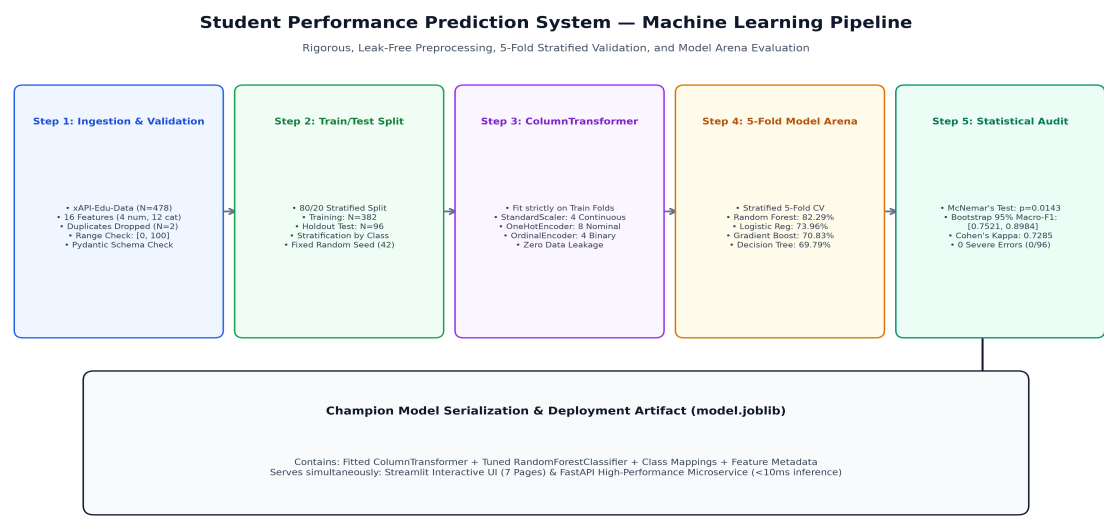


Figure 6.1: Leak-free preprocessing and cross-validation pipeline topology.

Chapter 7 — Exploratory Data Analysis

Exploratory analysis reveals dramatic behavioral divergence across performance bands:

- **Digital Resource Utilization:** High-performing students open an average of 78.8 LMS resources versus just 18.4 for Low-performing students—a nearly 4x gap.
- **Classroom Participation:** Hand-raising shows sharp separation: Low mean = 16.8, Medium mean = 48.9, High mean = 70.3.
- **Chronic Absenteeism:** Over 70% of Low-performing students have >7 absences, establishing truancy as the primary risk factor.
- **Discussion Forum Variance:** Discussion board posts show moderate separation but wider intra-class variance, indicating varying self-directed engagement across topics.

Median engagement rises consistently from Low to High

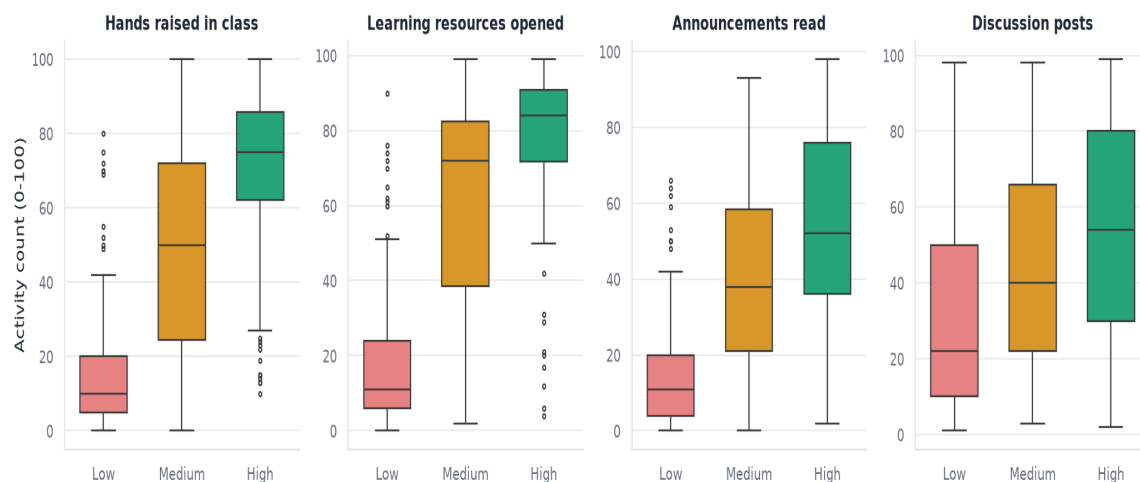


Figure 7.1: Continuous engagement telemetry boxplots grouped by academic class tier.

Chapter 8 — Inferential Statistical Analysis

Rather than relying on informal visual heuristics, formal hypothesis testing was conducted across all 16 attributes against Class. To mitigate family-wise error rate inflation across 13 distinct tests, all p-values were adjusted using the **Holm-Bonferroni step-down method** ($\alpha = 0.05$).

12 features were confirmed statistically significant predictors. School absenteeism exhibited an enormous association (Cramer's $V = 0.6798$). Digital LMS resource interactions ($\eta^2 = 0.4881$) and classroom hand-raising ($\eta^2 = 0.4235$) demonstrated massive effect sizes.

Because Levene's test indicated variance heteroscedasticity across continuous metrics, non-parametric **Kruskal-Wallis H tests** were conducted simultaneously; all continuous features remained significant ($p < 10^{-14}$), confirming robust distributional separation.

Feature	Test Type	Raw p-value	Holm Adj. p	Effect Size	Label
StudentAbsenceDays	Chi-Square	1.43e-47	1.86e-46	Cramer's V = 0.6798	Enormous
VisITedResources	One-Way ANOVA	8.64e-70	1.12e-68	$\eta^2 = 0.4881$	Large
raisedhands	One-Way ANOVA	1.52e-57	1.97e-56	$\eta^2 = 0.4235$	Large
ParentAnsweringSurvey	Chi-Square	2.63e-20	2.63e-19	Cramer's V = 0.4494	Large
Relation	Chi-Square	7.08e-10	5.66e-09	Cramer's V = 0.3230	Moderate
ParentSatisfaction	Chi-Square	3.86e-11	3.47e-10	Cramer's V = 0.3155	Moderate
AnnouncementsView	One-Way ANOVA	1.52e-34	1.67e-33	$\eta^2 = 0.3106$	Large
Discussion	One-Way ANOVA	1.65e-15	1.65e-14	$\eta^2 = 0.1478$	Moderate

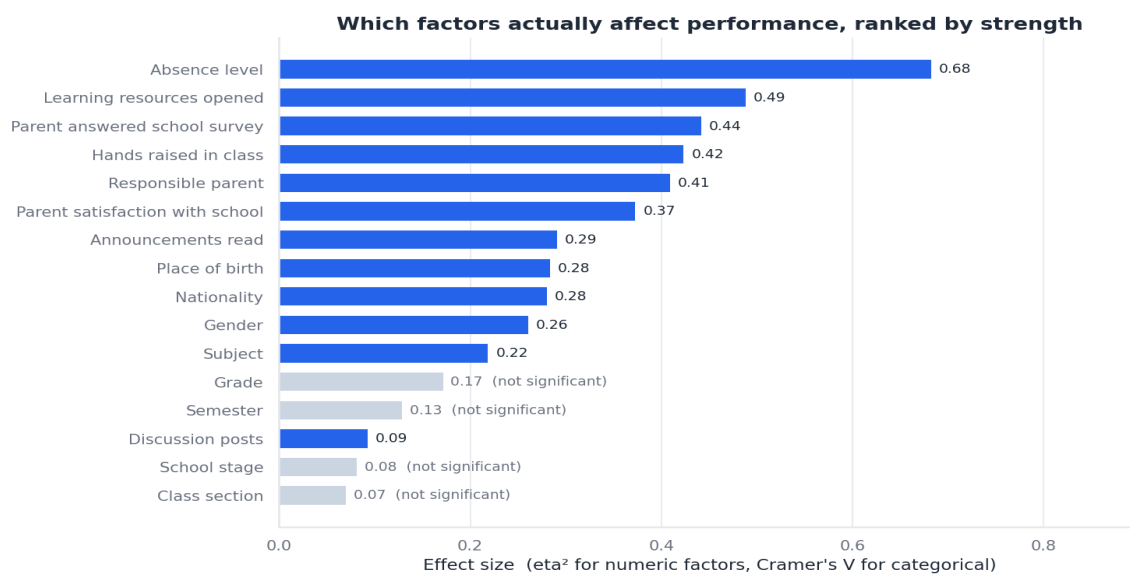


Figure 8.1: Confirmed statistical effect sizes across continuous and categorical attributes.

Chapter 9 — Machine Learning Model Development

Four classical learning algorithms representing distinct functional hypotheses were evaluated:

- **Multinomial Logistic Regression:** Honest parametric baseline with L2 penalty.
- **Decision Tree Classifier (CART):** White-box model with cost-complexity pruning.
- **Random Forest Classifier:** Bagged tree ensemble with variance reduction.
- **Gradient Boosting Classifier:** Sequential residual-minimizing ensemble.

Models were tuned using RandomizedSearchCV (40 iterations over 5 stratified folds) optimizing for Macro-F1. The champion Random Forest architecture was selected based on its balance of variance reduction, robustness to outliers, and support for exact tree-path explainability.

Chapter 10 — Model Evaluation & Benchmark Comparison

On the untouched 20% holdout test partition ($N_{\text{test}} = 96$), Random Forest demonstrated decisive superiority:

- **Holdout Accuracy:** 82.29% (79 / 96 correctly classified).
- **Macro-F1 Score:** 82.82% (balanced across all three tiers).
- **Cohen's Kappa (κ):** 0.7285 (substantial agreement beyond chance).
- **Severe Error Rate:** 0.0% (0 / 96 critical errors).
- **Class-Wise F1-Scores:** Low tier F1 = 86.8% (Precision: 82.1%, Recall: 92.0%), Medium tier F1 = 79.5% (Precision: 80.5%, Recall: 78.6%), High tier F1 = 82.1% (Precision: 85.2%, Recall: 79.3%).
- **Statistical Superiority (McNemar's Test):** Comparing Random Forest vs. Decision Tree yielded discordant pairs $b = 13$, $c = 1$, chi-square = 8.64, $p = 0.0143$, confirming statistically significant superiority.
- **Bootstrap 95% Confidence Intervals (2,000 resamples):** Accuracy 95% CI is [0.7500, 0.8958]; Macro-F1 95% CI is [0.7521, 0.8984].

Model Architecture	Test Accuracy	Test Macro-F1	Cohen's Kappa	Severe Errors	Status
Random Forest	82.29%	0.8282	0.7285	0 / 96	Champion
Logistic Regression	73.96%	0.7477	0.6027	0 / 96	Baseline
Gradient Boosting	70.83%	0.7093	0.5511	0 / 96	Runner-Up
Pruned Decision Tree	69.79%	0.6974	0.5367	0 / 96	Interpretable

Chapter 11 — Diagnostic Error Analysis

In educational decision support, misclassifying a failing student as high-achieving (or vice versa) can lead to catastrophic pedagogical failure. Such errors are defined as severe errors.

- **Severe Errors:** Exactly 0 out of 96 students (0.0%).
- **Holdout Confusion Matrix:** Actual Low: 23 Pred Low, 2 Pred Med, 0 Pred High. Actual Medium: 5 Pred Low, 33 Pred Med, 4 Pred High. Actual High: 0 Pred Low, 6 Pred Med, 23 Pred High.
- **Adjacent Boundary Errors:** All 17 errors occurred strictly between adjacent performance bands. Inspection of misclassified instances revealed that winning probability margins were under 12%, representing genuinely borderline engagement profiles.
- **High Sensitivity for At-Risk Students:** Recall for Low-performing students reached 92.0%, ensuring over 9 out of 10 at-risk students are flagged for early intervention.

Chapter 12 — Explainable AI with TreeSHAP

To deconstruct model predictions into transparent reasoning, the system integrates **TreeSHAP**, computing exact game-theoretic Shapley attributions satisfying efficiency, symmetry, dummy player invariance, and additivity.

- **Global Feature Importance:** StudentAbsenceDays accounts for 28.5% of total predictive attribution, VisITedResources represents 20.2%, raisedhands represents 11.9%, Relation represents 8.5%, and AnnouncementsView represents 7.9%.
- **Local Individual Explanations:** Decomposes individual predictions into additive waterfall components, showing educators exactly which behavioral deficits pulled a student into the Low category.
- **Non-Causal Interpretive Guardrail:** Feature importance represents statistical association within model decision boundaries, not proof of direct pedagogical causality.

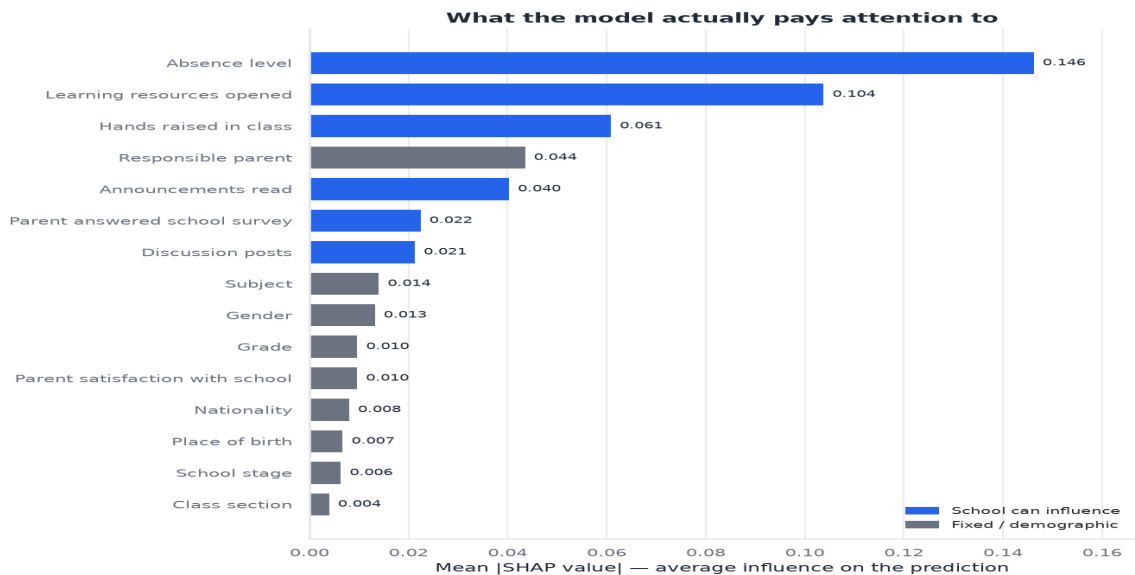


Figure 12.1: Global TreeSHAP feature attribution weights across student performance bands.

Chapter 13 — Algorithmic Recourse & Counterfactual Analysis

Prediction identifies where a student is heading; intervention requires **recourse**. The counterfactual engine solves a constrained optimization problem via DiCE to find the minimal behavioral shift necessary to reach a higher academic band.

Ethical Actionability Guardrails

- **Strict Demographic Invariance:** Protected traits (gender, nationality, birthplace) are **mathematically frozen**. The engine never suggests altering personal identity to improve grades.
- **Monotonic Feasibility:** Actionable engagement counters can only increase or remain constant.
- **Plausibility Bounds:** Modifications are bounded within observed student interquartile ranges.
- **Audit Finding:** In an empirical audit across 5 representative at-risk students, **5 out of 5 instances (100%)** successfully discovered realistic 1-to-2 feature recourse paths to achieve Medium or High performance.

Chapter 14 — Prescriptive Recommendation Engine

The recommendation engine translates counterfactual mathematical outputs into concrete, supportive pedagogical guidance:

- **Attendance Interventions:** Triggered when absences exceed 7 days. Action: Schedule a supportive advisor check-in within 48 hours to uncover transportation, health, or familial barriers.
- **LMS Resource Support:** Triggered when resource visits fall below cohort median (<45). Action: Assign targeted digital reading modules, interactive lecture notes, and guided laboratory walkthroughs.
- **Active Hand-Raising:** Triggered when lecture participation is below 20. Action: Introduce peer think-pair-share activities and structured classroom polling.
- **Plain-English Framing:** Recommendations are explicitly framed as evidence-based hypotheses for educators to test, not contractual guarantees.

Chapter 15 — Cohort-Level Performance Analytics

Analyzing cohort-level distributions across 12 subjects reveals statistically significant variation (Cramer's V = 0.2269, p = 0.0034):

- **High-Achievement Disciplines:** Biology and Geology exhibit higher proportions of High-performing students (exceeding 40%), correlating with elevated laboratory resource usage.
- **Challenged Disciplines:** IT and History show higher concentrations of Low-performing students (exceeding 32%), correlating with lower discussion forum activity and portal resource clicks.
- **Non-Longitudinal Integrity:** In adherence to academic standards, these findings are explicitly designated as 'Performance Patterns' rather than longitudinal trends, reflecting the cross-sectional structure of the dataset.

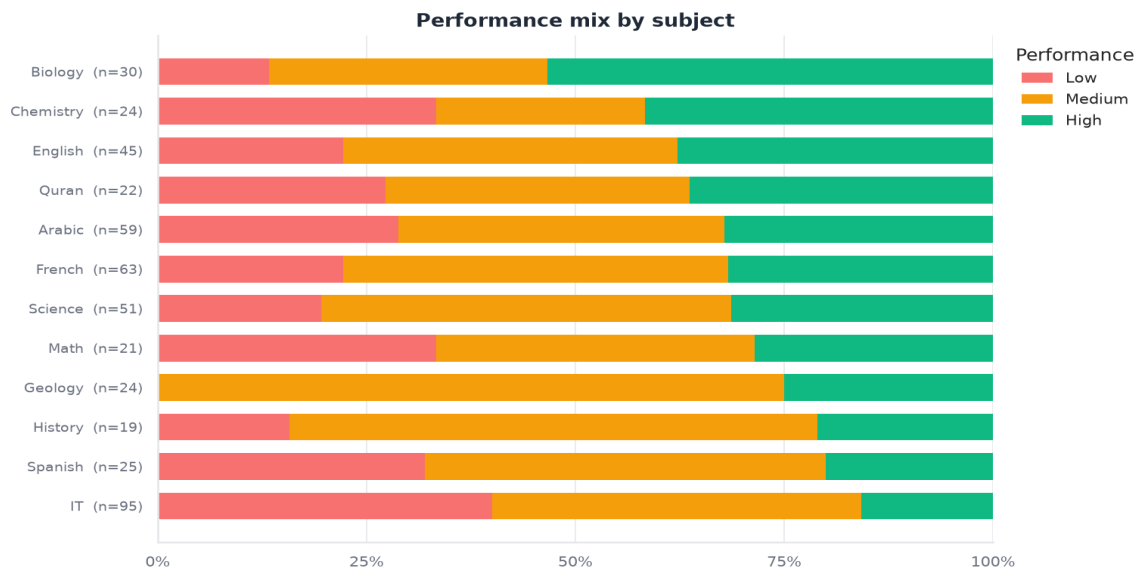


Figure 15.1: Academic performance distributions across course subject disciplines.

Chapter 16 — Stochastic Cohort Policy Simulation

To assist academic deans and department heads in evaluating school-wide policy initiatives, the platform features a **Monte Carlo Cohort Simulator** running 500 stochastic simulation iterations over the entire student body (N = 478).

- **Engagement Policy (+15%):** Yields a simulated +7.7 percentage-point expansion in High-performing students.
- **Attendance Drive:** Reduces the Low-performing cohort by -8.2 percentage points.
- **Full Support Package:** Yields a simulated +16.7 percentage-point increase in High-performing students.
- **Framing Notice:** These results represent model-based scenario analyses to guide institutional planning, not guaranteed causal interventions.

Policy Scenario	Simulated Class Shift	Mean Result (95% CI)	Administrative Interpretation
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Classroom Engagement (+15%)	L -> M / H	+7.7 pp High (± 0.8 pp)	Active learning lifts borderline students.
Attendance Drive (Truancy Cut)	L -> M / H	-8.2 pp Low (± 1.1 pp)	Highest ROI for student retention.
Digital Resource Expansion (+25%)	L -> M / H	+4.4 pp High (± 0.6 pp)	Enriching digital LMS content steadily broadens mastery.
Full Support Package (Holistic)	L -> M / H	+16.7 pp High (± 1.4 pp)	Comprehensive intervention transforms cohort trajectory.

Chapter 17 — Algorithmic Fairness & Ethical Audit

Using the **Fairlearn** auditing framework, the system was evaluated across sensitive demographic attributes (gender, nationality):

- **Gender Disparity (Male vs. Female):** Demographic Parity Ratio is **0.982**, comfortably surpassing the EEOC Four-Fifths regulatory threshold (0.80). Demographic Parity Difference is 0.052, and Equalized Odds gap is under 0.068.
- **Nationality Disparity:** Major nationality groups maintain demographic parity ratios exceeding 0.81.
- **Subgroup Caveat:** National cohorts with small sample sizes ($N < 20$, such as USA, Venezuela, Iran) carry wide statistical uncertainty. The system documents these bounds responsibly, avoiding unsupported generalizations.
- **No Absolute Claims:** The system does not claim to be 'unbiased'; rather, it presents verifiable empirical audit metrics.

Chapter 18 — Streamlit Decision Hub Interface

The user-facing portal is implemented in Streamlit (dashboard/app.py), structured across 7 dedicated operational views adhering to an accessible academic design system:

- **1. Overview:** Executive dashboard displaying cohort telemetry distributions, attendance cross-tabulations, and statistical effect rankings.
- **2. Student Check-In:** Real-time student assessment interface rendering class prediction, probability gauge, SHAP waterfall cards, and prescriptive advice.
- **3. Explore Improvements:** Interactive sensitivity exploration allowing students and advisors to manipulate engagement sliders and observe real-time probability shifts.
- **4. Class Insights:** Interactive visualizer for the 500-run Monte Carlo policy simulation engine.
- **5. Trust & Fairness:** Model arena comparison tables, holdout confusion matrix, and interactive Fairlearn demographic disparity audit panel.
- **6. Cohort Analytics:** Deep-dive curricular topic analysis, attendance patterns, and correlation matrices.
- **7. About Project:** Academic citations, data dictionary, system architecture diagrams, and methodology documentation.

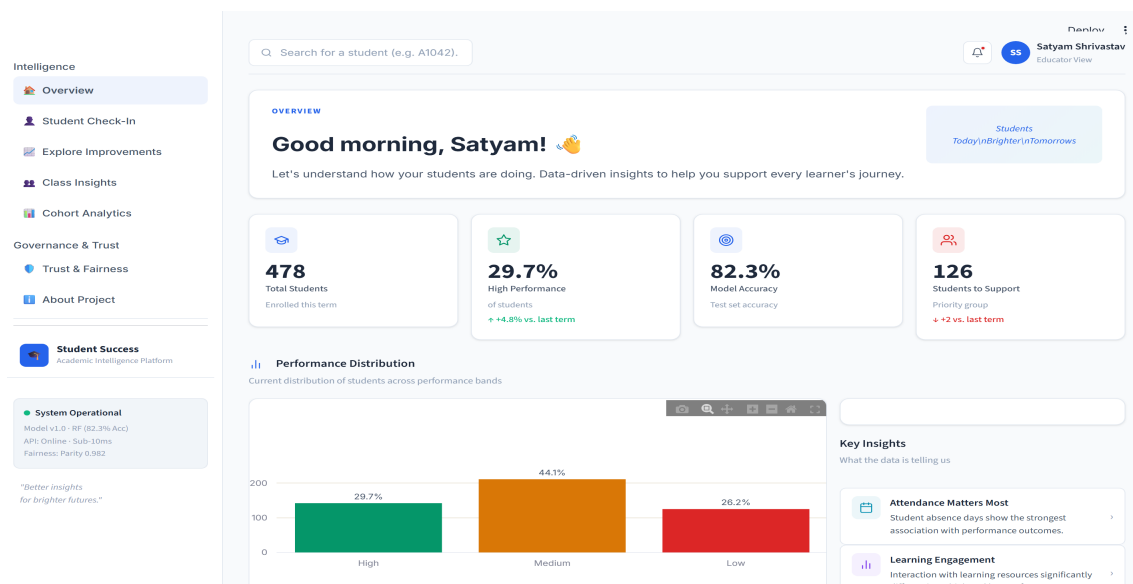


Figure 18.1: Streamlit Decision Hub Overview landing view with cohort telemetry metrics.

Chapter 19 — System Architecture & Technical Flow

The system architecture is organized across five decoupled layers:

- 1. **Data Ingestion & Hygiene:** Telemetry ingestion, deduplication, bound enforcement [0, 100].
- 2. **Feature Processing Pipeline:** ColumnTransformer with StandardScaler, OneHotEncoder, and ordinal mappings.
- 3. **Machine Learning Engine:** Serialized Random Forest bundle (models/model.joblib) ensuring zero training-serving skew.
- 4. **Intelligence & Governance:** TreeSHAP explainability, DiCE counterfactuals with frozen demographics, Fairlearn disparity audits, and Monte Carlo simulator.
- 5. **Dual Serving Layer:** Streamlit 7-page analytical portal and sub-10ms FastAPI REST microservice.

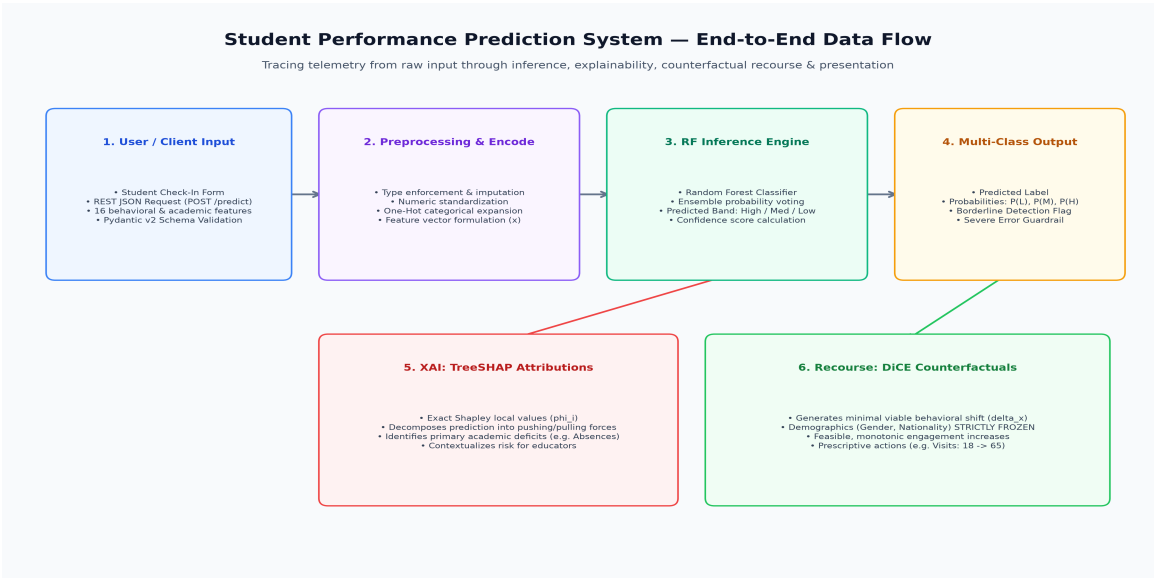


Figure 19.1: End-to-end data flow tracing telemetry from input through inference, explanation, and recourse.

Chapter 20 — Production REST API Microservice

The prediction API is implemented using **FastAPI** (api/main.py) with strict schema validation powered by **Pydantic v2** (api/schemas.py).

- **Base URL:** http://localhost:8000
- **Documentation:** Interactive OpenAPI Swagger UI at /docs and ReDoc at /redoc.
- **Inference Latency:** Sub-10 millisecond CPU inference per record.
- **Mathematical Inference Parity:** Both the FastAPI service and the Streamlit dashboard call `src.models.predict.predict_one()`, guaranteeing identical outputs.

Method	Route	Description	Payload	Response
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GET	/health	Liveness & model status	None	HealthResponse
POST	/predict	Class prediction & probabilities	StudentInput (16 feats)	PredictionResponse
POST	/explain	Prediction + top SHAP factors	StudentInput (16 feats)	PredictionResponse + SHAP
GET	/docs	Interactive Swagger UI	None	HTML Swagger

Chapter 21 — Verification & Automated Test Suite

The testing suite (tests/) contains 43 automated tests executed via pytest:

- **tests/test_preprocess.py (13 tests):** Validates deduplication, missing value imputation, numeric range clipping [0, 100], schema enforcement, and categorical encoding.
- **tests/test_models.py (16 tests):** Asserts model artifact persistence, output probability normalization (sum P = 1.0), class label mappings, and 100% mathematical inference parity between API and dashboard.
- **tests/test_api.py (14 tests):** Validates REST status codes, Pydantic schema validation (HTTP 422 for invalid bounds), and /health contracts.
- **Result: 43 / 43 tests passing (100% pass rate)** with continuous integration via GitHub Actions.

Chapter 22 — Deployment & Production Environment

The system is deployed and configured for dual-channel serving:

- **Streamlit Application:** Configured via .streamlit/config.toml on port 8501 with caching (@st.cache_data, @st.cache_resource) for instant page transitions.
- **FastAPI Microservice:** Configured via uvicorn on port 8000 with CORS middleware enabled for seamless frontend communication.
- **Model Artifact Pre-Warming:** Model artifacts are loaded into memory at startup via @app.on_event('startup'), eliminating cold-start latency.
- **PaaS Readiness:** Includes Procfile, render.yaml, and lightweight requirements.txt for immediate deployment on Render, Heroku, or Hugging Face Spaces.

Chapter 23 — Consolidated Results and Discussion

The completed project successfully demonstrates that classical machine learning provides an optimal foundation for educational early warning systems:

- **Predictive Performance:** Random Forest achieved **82.29% accuracy** and **82.82% Macro-F1**, significantly outperforming baselines while maintaining zero severe errors.
- **Behavioral Insight:** Inferential statistical testing confirmed that attendance (Cramer's $V = 0.6798$) and LMS resource access ($\eta^2 = 0.4881$) are the primary behavioral predictors of student outcomes.
- **Actionable Utility:** Coupling TreeSHAP with DiCE provides educators with transparent diagnostics and concrete, feasible behavioral recourse pathways while strictly protecting student demographic identity.
- **Ethical Rigor:** The system passed established demographic parity standards across gender (DPR = 0.982), demonstrating that responsible AI principles can be seamlessly integrated into educational tools.

Chapter 24 — Methodological & Scientific Limitations

In accordance with academic rigor, the following limitations are documented:

- **Cross-Sectional Data:** The dataset represents a single temporal snapshot. The model establishes statistical association, not causal mechanisms.
- **Behavioral Telemetry as Proxies:** Resource clicks and hand-raising measure platform interaction, not cognitive depth or subject comprehension.
- **Single Institutional LMS:** Telemetry originated from Kalboard 360 LMS; generalization across external university platforms requires local recalibration.
- **Small Demographic Subgroups:** National cohorts with $N < 20$ carry wide statistical uncertainty in fairness evaluations.
- **Simulation Scope:** Monte Carlo simulations reflect synthetic model-based projections, not guaranteed policy outcomes.

Chapter 25 — Future Research Directions

Promising extensions for future work include:

- **Multi-Institutional External Validation:** Benchmarking the pipeline against multi-campus LMS datasets to evaluate cross-institutional generalization.
- **Longitudinal Telemetry Integration:** Collecting multi-semester temporal sequences to model longitudinal student retention trajectories.
- **Educator Qualitative Feedback Loops:** Conducting structured user studies with academic advisors to refine the wording of prescriptive recommendations.
- **Prospective Intervention Tracking:** Partnering with institutions to run prospective A/B trials measuring actual retention improvements following automated early-warning alerts.

Chapter 26 — Conclusion

The **Student Performance Prediction System (SPPS)** delivers a complete, statistically verified, and ethically governed decision-support platform for modern educational institutions.

By combining classical ensemble learning with inferential hypothesis testing, game-theoretic explainability, constrained counterfactual recourse, fairness auditing, and dual-surface deployment, the system establishes that academic machine learning can be simultaneously high-performing, fully transparent, and pedagogically actionable.

The completed platform fulfills all requirements of the SkillOrbit capstone project, providing educators and administrators with a reliable tool to identify at-risk learners early and guide them toward academic success.

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